YONGGANG JIANG

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PROFESSIONAL EXPERIENCE

ETH Zürich *Oct. 2026 – Now*
Junior Fellow at the ETH Institute for Theoretical Studies

EDUCATION

Max Planck Institute for Informatics *Oct. 2021 – Sep. 2026*
PhD in Computer Science
Advisor: Danupon Nanongkai

University of California, Berkeley *Jan. 2020 – May 2020*
Berkeley International Study Program
GPA: 4.0/4.0

Nanjing University *Sep. 2017 – Jul. 2021*
B.S. in Computer Science and Technology
GPA: 94.4/100, Ranking: 1/31

HONORS & AWARDS

Google PhD Fellowship	*2025*
Distinguished Paper Award, SPAA 2025	*2025*
Outstanding Graduate, Nanjing University	*2021*
National Elite Program Scholarship, Outstanding Prize (top 1)	2018, 2019, 2020
China Collegiate Programming Contest (CCPC), Gold Medal	*2018*
National Olympiad in Informatics in Provinces (NOIP), First Prize	*2015*

RESEARCH INTERESTS

I work broadly on algorithms and optimization. My goal is to design algorithms that are theoretically efficient and fast in practice. I am particularly interested in:

- Graph algorithms
- Parallel computing
- Algorithm engineering
- AI for mathematics and optimization

I am currently developing LLM harnesses for mathematical research. I lead the development of TCS-prover and contribute to ProofCouncil.

SELECTED PUBLICATIONS

* Authors are listed alphabetically with **equal contribution** (see convention in mathematics).

Published

[1] **Almost Optimal Streaming and Cut-Query Algorithms for Undirected Max-Flow**
Yonggang Jiang, Tatsuya Terao.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2027.

[2] **Combinatorial Minimum Cost Flow in Almost-Linear Time on Dense Graphs**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
IEEE Symposium on Foundations of Computer Science (FOCS) 2026.

[3] **Parallel Small Vertex Connectivity in Near Linear Work and Polylogarithmic Depth**
Yonggang Jiang, Changki Yun.
IEEE Symposium on Foundations of Computer Science (FOCS) 2026.

[4] **Deterministic Negative-Weight Shortest Paths in Nearly Linear Time via Path Covers**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
ACM Symposium on Theory of Computing (STOC) 2026.

[5] **DAG Projections: Reducing Distance and Flow Problems to DAGs**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
ACM Symposium on Theory of Computing (STOC) 2026.

[6] **Reviving Thorup's Shortcut Conjecture**
Aaron Bernstein, Henry Fleischmann, George Z. Li, Bernhard Haeupler, Gary Hoppenworth, Yonggang Jiang, Maximilian Probst Gutenberg, Seth Pettie, Thatchaphol Saranurak, Leon Schiller.
ACM Symposium on Theory of Computing (STOC) 2026.

[7] **Perfect Simulation of Las Vegas Algorithms via Local Computation**
Xinyu Fu, Yonggang Jiang, Yitong Yin.
Innovations in Theoretical Computer Science (ITCS) 2026.

[8] **Reducing Shortcut and Hopset Constructions to Shallow Graphs**
Bernhard Haeupler, Yonggang Jiang, Thatchaphol Saranurak.
SIAM Symposium on Simplicity in Algorithms (SOSA) 2026.

[9] **Directed and Undirected Vertex Connectivity Problems are Equivalent for Dense Graphs**
Olivier Fischer, Yonggang Jiang, Sagnik Mukhopadhyay, Sorrachai Yingchareonthawornchai.
SIAM Symposium on Simplicity in Algorithms (SOSA) 2026.

[10] **Minimum s-t Cuts with Fewer Cut Queries**
Danupon Nanongkai, Yonggang Jiang, Pachara Sawettamalya.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.

[11] **Shortcuts and Transitive-Closure Spanners Approximation**
Parinya Chalermsook, Yonggang Jiang, Sagnik Mukhopadhyay, Danupon Nanongkai.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.

[12] **New Oracles and Labeling Schemes for Vertex Cut Queries**
Yonggang Jiang, Merav Parter, Asaf Petruschka.
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2026.

[13] **Parallel $(1+\epsilon)$-Approximate Multi-Commodity Mincost Flow in Almost Optimal Depth and Work**
Bernhard Haeupler, Yonggang Jiang, Yaowei Long, Thatchaphol Saranurak, Shengzhe Wang.
IEEE Symposium on Foundations of Computer Science (FOCS) 2025.